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1) Define Data Mining and list any two basic data mining tasks.

Explain the steps involved in the KDD Process and justify the role of data preprocessing.

Ans

Data Mining:-

Data Mining is the process of discovering useful patterns, relationships and knowledge from large datasets.

Two basic data mining tasks:

1. Classification

2. Clustering

Steps in KDD Process:

1. Data Selection

2. Data Cleaning

3. Data Integration

4. Data Transformation

5. Data Mining

6. Pattern Evaluation

7. Knowledge Presentation

2) What is Data Cleaning? Mention two common data quality issues.

Explain methods to handle missing values and noisy data with suitable examples.

Ans

Data cleaning:

Data cleaning is the process of detecting and correcting inaccurate or incomplete data.

Two data quality issues:

* Missing values.

* Noisy or inconsistent data

Handling missing values:

* Replace with mean/median/mode.

* Predict values using machine learning

Example: missing age $\rightarrow$ replace with average age.

Handling Noisy Data:

- * Binning
- * Regression
- * Clustering

3) Define Data Integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled.

Ans Data integration:

Data Integration combines data from multiple sources into a single unified dataset.

Challenges:

- 1) Data redundancy.
- 2) Data inconsistency

Handling:

- * Remove duplicate records.
- * Standardize formats and resolve conflicting values using data cleaning.

4) what is Covariance Analysis? List the two covariance formulas.

Ans Covariance Analysis

Covariance measure how two variables change together

Population Covariance:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance:

$$\text{Cor}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Interpretation:

- * Positive $\rightarrow$ increase together.
- * Negative $\rightarrow$ One increases while the other decreases
- * Zero - no linear relationship.

Suppose the two stocks A and B have the following values: (2,5), (3,8), (5,10), (4,11), (6,14)

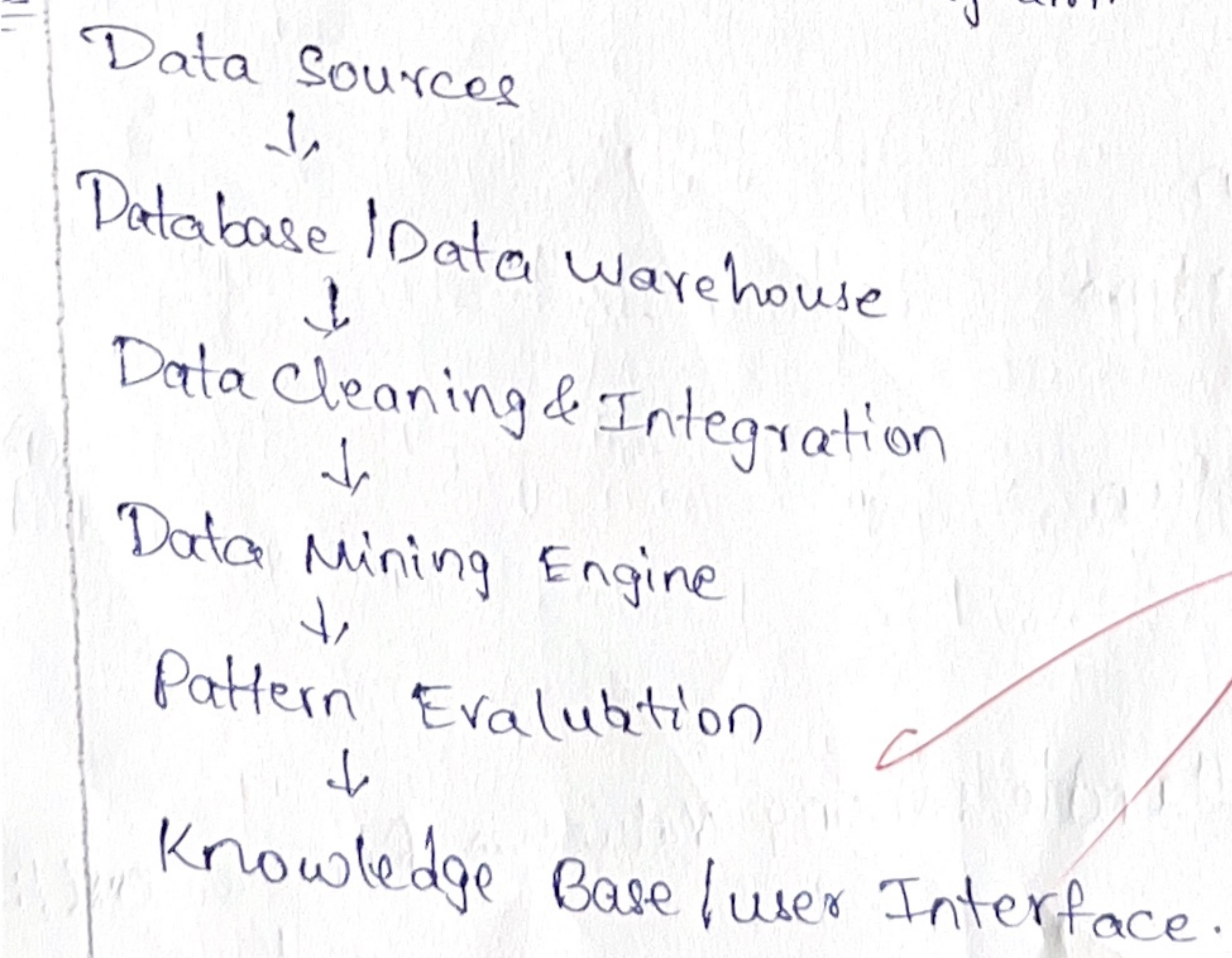
Mean of A = 4

Mean of B = 9.6

$$\text{Covariance} = \frac{21}{4} = 5.25$$

Since Covariance is Positive, both stocks generally rise together due to similar industry trends.

Explain the architecture of a typical Data Mining System with a neat diagram.



Components:

- * Data Sources - collect raw data
- * Database / Data warehouse - stores data
- * Data Cleaning - improves data quality
- * Data Mining Engine - discovers patterns

7) Find Covariance for:

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

Ans

$$\text{Mean of } x = 6.2$$

$$\text{Mean of } y = 5.8$$

$$\text{Covariance} = \frac{-2.8}{4} = -0.7$$

Since Covariance is negative, the variables have a weak inverse relationship.

8) Describe the major steps in Data preprocessing. Propose a preprocessing Pipeline.

Ans

Major steps:

- 1) Data Cleaning
- 2) Data Integration
- 3) Data Transformation
- 4) Data Reduction
- 5) Data Discretization

Pipeline:

Raw Data $\rightarrow$ Handle Missing Values $\rightarrow$ Remove outliers

↓
Standardize formats
← Normalize Data
← Reduce Data
← Prepared Data

This sequence ensures high-quality data before mining.

Form Equal-Frequency (Equi-depth) Binning

data set:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

total values = 27

Choose 3 bins

Each bin = 9 values

Bin 1:

13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2:

22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3:

35, 35, 35, 36, 40, 45, 46, 52, 70

This method reduces noise while keeping an equal number of values in each bin.

Q) Explain Data Discretization compare Equal-width and Equal-frequency Binning.

Data Discretization:

It converts continuous numerical values into intervals (bins) making analysis easier.

Equal-width

- * Equal interval size
- * Sensitive to outliers
- * Simple to implement

Equal Frequency

- * Equal number of values
- * Less affected by outliers
- * Better for uneven data

Importance:

- * Reduce noise
- * Improves mining efficiency
- * Simplifies data analysis
- * Enhances model performance